

Teaching Inflation Expectations with Professional Forecast Data and a Classroom Experiment

Brian C. Jenkins*

May 19, 2026

Abstract

Inflation expectations appear throughout undergraduate economics including in the Fisher equation, models of aggregate supply based on the Phillips curve, and discussions of monetary policy. However, despite advances in the empirical and theoretical literatures on inflation expectations, undergraduate courses continue to present expectations as exogenous or rational, leaving students with little appreciation of how they are formed or measured. This paper provides instructors with a concise guide to these developments and offers ways to integrate them into teaching. I show how data from the Federal Reserve Bank of Philadelphia's Survey of Professional Forecasters (SPF) can be used to illustrate empirical facts and motivate classroom discussions about expectation formation. I provide examples of lecture applications, suggested homework assignments using SPF data, and an in-class inflation-forecasting exercise that helps students connect theoretical concepts to real-world data.

1 Introduction

In this paper, I describe two sources of data on the public's inflation expectations, the Survey of Professional Forecasters (SPF) and the Michigan Survey of Consumers, and I provide a ready-to-use dataset that can serve as the basis for lecture content and homework assignments. The dataset includes professional inflation forecast data, realized inflation, and nominal interest rates, making it straightforward to construct an *ex ante* real interest rate series, a quantity that is central to macroeconomic models but rarely matched to data in textbooks. I also describe an inflation forecasting experiment in which students are given recent historical U.S. macroeconomic data and asked

*Department of Economics, University of California, Irvine, Phone: +1 (949) 824-0640, Fax: +1 (949) 824-2182, Mailing address: 3151 Social Science Plaza, Irvine, CA 92697-5100, Email: bcjenkin@uci.edu.

to forecast inflation one year ahead. The experiment creates a natural bridge to the professional forecasting data: students will find that their forecasts are broadly comparable to those of professionals, which helps give empirical credibility to the assumption that the public forms reasonable inflation expectations.

There is well-documented evidence that professional forecasters produce systematic forecast errors (Angeletos, Huo and Sastry 2021). It's worthwhile for instructors who teach macroeconomics to be informed about how inflation expectations are formed in practice. After working with the dataset described in Section 2, students should be able to discuss the similarities and differences between forecasted and actual inflation and to compute an *ex ante* real interest rate using survey forecast data. After completing the experiment described in Section 3, students should understand that inflation is more difficult to forecast when it is more variable and that forecast errors tend to be serially correlated.

The value of classroom experiments in economics instruction has long been appreciated (Holt 1999). Traditionally experiments have been run in-class, but online experiments are gaining popularity with platforms like MobLab, classEx, and VeconLab (Atwood, Emerson, Knox and Taznin 2023). Carter and Emerson (2012) provide evidence that online experiments are as effective as in-class experiments. While classroom experiments in microeconomics appear to be more common, a number of experiments and games have been developed for macroeconomics, including experiments on money demand (Ewing, Kruse and Thompson 2004, Chen 2018), monetary policy (Duffy and Jenkins 2019), and the Lucas islands model of imperfect information (Hashimzade, Kirsanov and Kirsanova 2026). The oTree platform has made it easier for instructors to develop their own experiments (Chen, Schonger and Wickens 2016). Cortés, Mantilla and Prada (2023) use it to develop a web-based game on the dynamic management of a renewable resource. To my knowledge, no classroom experiment focused specifically on inflation forecasting has been discussed in the literature and the experiment described in Section 3 fills that gap.

2 Inflation Forecast Data

In this section I describe two sources of inflation forecast data and provide a ready-to-use dataset combining inflation forecast data and macroeconomic data from FRED. I show that the inflation forecast data can be used to construct an *ex ante* real interest rate. Finally, I highlight two teachable facts that arise from the data. First, professional inflation forecasts broadly move with inflation over long time horizons. Second, in the short-run, inflation forecast errors show strong positive correlation, implying that professionals often systematically over- or under-forecast inflation for many consecutive dates.

2.1 Sources

The Survey of Professional Forecasters (SPF) is a quarterly survey of U.S. macroeconomic forecasts reported by the Federal Reserve Bank of Philadelphia.¹ Begun in 1968 as a collaboration of the American Statistical Association and the National Bureau of Economic Research, the Philadelphia Fed assumed responsibility for the SPF in 1990. The SPF covers a broad range of U.S. macroeconomic variables, including output, inflation, labor market conditions, and interest rates. For most series, the SPF provides the forecasts of individual forecasters, the mean and median forecast, and a measure of the cross-sectional dispersion of forecasts. Therefore it's easy to investigate the central tendency of forecasts and the degree of disagreement across forecasting professionals.

The SPF reports forecasts for three inflation measures: the GDP deflator, the CPI, and the PCE deflator. GDP deflator forecasts are reported in levels of the price index itself and so implied inflation forecasts must be computed. CPI and PCE forecasts are reported directly as inflation rates. The GDP deflator forecast series is the longest, extending back to 1968 while CPI forecasts begin in 1981 and PCE forecasts only in 2007. While CPI and PCE inflation are more prominent in public discourse on inflation and monetary policymaking, the GDP deflator series is preferable here because its range

¹<https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/survey-of-professional-forecasters>

spans the high and variable inflation of the 1970s. Figure 1 shows the annual GDP deflator inflation rate and the one-year-ahead median SPF forecast.

An alternative to the SPF is the Michigan Survey of Consumers produced by the Survey Research Center at the University of Michigan.² It is a nationally representative monthly survey of approximately 1,000 U.S. households. The survey covers consumer attitudes including their expectations for inflation over the next year. In contrast with the forecasters who are represented by the SPF, respondents in the Michigan Survey are households, not professionals, and so the data arguably reflect a more accurate view of the general public's expectations. The data are available both from the survey website at monthly, quarterly, and annual frequencies and from FRED at the monthly frequency.³ Figure 2 shows the annual CPI inflation rate and the one-year-ahead median Michigan Survey inflation forecast.

The Michigan Survey reports inflation expectations dating to the early 1960s, but observations are irregular and regular monthly coverage begins only in 1978. As a result, the forecast data in the Michigan Survey does not fully cover the high and variable inflation of the 1970s. The GDP deflator forecast data from the SPF is my preferred inflation forecast series for teaching because observations go back to 1970.

2.2 Dataset for Teaching

To make it easier for instructors to incorporate SPF inflation forecast data into their courses, I provide a dataset that combines SPF forecast data with actual inflation and interest rate data from FRED into a single file. The file, along with the code to produce it, is available at the GitHub repository for this project.⁴ The dataset eliminates the need for instructors or students to download and merge data from multiple sources, and is suitable both as a basis for lecture content and for empirical assignments. It contains three variables at both quarterly and annual frequencies: the median one-year-ahead GDP deflator inflation forecast from the SPF, realized GDP deflator inflation computed

²<https://www.sca.isr.umich.edu/>

³<https://fred.stlouisfed.org/series/MICH>

⁴<https://github.com/letsgoexploring/teaching-inflation-expectations>

as the year-over-year percentage change in the GDP deflator (GDPDEF from FRED), and the one-year Treasury rate (GS1 from FRED). The interest rate is included so that users can easily compute the ex ante real interest rate (see Section 2.4).

2.3 Median Forecasts and Forecast Errors

Figure 3 shows the median SPF forecast of GDP deflator inflation alongside actual inflation over the full sample period. Shading between the lines indicates periods when actual inflation exceeded the median forecast and periods when it fell short. Several features are worth highlighting in the classroom. First, the median forecast broadly tracks actual inflation over long periods of time. As inflation rose in the 1970s so did inflation forecasts and as inflation fell in the 1980s and 90s, forecasts gradually followed suit. Second, the forecast errors are systematic and do not appear to be fully random. Forecast errors are consistently positive or negative for consecutive years, suggesting that professional forecasters are slow to update their beliefs in response to new information. The shading makes this pattern visually immediate: rather than alternating randomly, blue and red regions cluster, with long stretches of systematic over- or underforecasting. This observation presents a teachable moment. The persistence of forecast errors in one direction is consistent with a common assumption of short-run models: inflation expectations are approximately constant in the short-run.

The data described in this section allow instructors to give students a perspective on what professional forecasters produce. In Section 3, I complement this with a classroom experiment that places students in the position of the forecaster, giving them direct experience of the difficulty of forecasting inflation in real time. Among other things, they learn that inflation is harder to forecast when it is more variable and that forecast errors are often serially correlated.

2.4 The Ex Ante Real Interest Rate

The Fisher equation defines the ex ante real interest rate as the difference between the nominal interest rate and expected inflation:

$$r_t = i_t - E_t\pi_{t+1} \tag{1}$$

While the ex ante real interest rate is an important endogenous variable in many macroeconomic models, computing it requires taking a stand on expected inflation. The dataset described in the previous section makes this straightforward: students can compute the real interest rate at each date by subtracting the inflation forecast column from the nominal interest rate column.

Figure 4 illustrates the result. The top panel shows the one-year nominal interest rate alongside the median SPF forecast of GDP deflator inflation. The bottom panel shows the implied ex ante real interest rate. During the high-inflation period of the 1970s, nominal rates were elevated but expected inflation was also high, leaving the real rate low or negative. The Volcker disinflation of the early 1980s appears not merely as a rise in nominal rates but as a sharp increase in the real rate, as nominal rates rose faster than expectations adjusted. The real rate declined over the 1980s and 1990s, but more slowly than the nominal rate. Following the Great Recession, the nominal rate was held near zero for roughly half a decade, pushing the real rate below zero for much of that period.

3 Experiment Design and Implementation

I describe an inflation forecasting experiment that gives students practical experience forecasting inflation under uncertainty. Students' experience with the experiment creates a bridge for them to make sense of the forecasting data from the SPF. The experiment takes about 20 minutes to run and can be done outside of class. Debriefing the experiment takes about 5 to 10 minutes of class time.

3.1 Experimental Overview

The experiment consisted of the following sequence: an introduction page, two instruction pages, 20 experimental rounds, a post-experiment survey, and finally a results page. Students were given 20 minutes to complete the entire experiment, from introduction to final results. The first two rounds of the experiment were practice rounds to give the students a chance to get used to the interface. The remaining 18 rounds were scored according to the procedure described below.

Each round of the experiment corresponded to one year of U.S. macroeconomic history from 1970–1989. However, students were not shown the calendar years corresponding to each round because many might be aware that the United States experienced high and variable inflation during the 1970s. Figure 5 shows a typical decision screen. On each decision screen, students were shown a table containing data from the current period and the previous four. Data provided included the inflation rate, the unemployment rate, and the federal funds rate. Students were also shown a line chart of the inflation rate along with their past forecasts so they could see their forecast errors. Students entered a numeric forecast constrained to the interval $[-5.0\%, 15.0\%]$ to prevent wildly incorrect forecasts.

After each submission, students immediately received feedback showing their forecast, the actual inflation rate for that period, their forecast error, their squared error for that round, and their running mean squared error (MSE) across all scored rounds completed to that point, along with an indication of whether they had over- or under-forecast. Figure 6 depicts the feedback screen.

The experiment was developed and implemented using oTree 6.0.10. The interface was optimized for use with both desktop and mobile devices, since many students would be using phones or tablets. The experiment was deployed to the students using a Heroku server. The code for the experiment and instructions is available on the GitHub site for this project.

3.2 Data

The experiment presented students with U.S. annual macroeconomic data from the 20 year period 1970–1989. This period was selected because it covers the high and variable inflation of the 1970s and the subsequent disinflation of the 1980s. Data were obtained from the Federal Reserve Economic Data (FRED) database. The inflation rate was constructed as the annual percentage change in the Personal Consumption Expenditures (PCE) price index (DPCERG3A086NBEA). I used PCE deflator inflation instead of the GDP deflator because it’s the Fed’s preferred measure of inflation and so as much as possible I try to use that measure in my class. The unemployment rate was computed as the annual mean of the monthly values (UNRATE). The federal funds rate was computed as the annual mean of the monthly series (FEDFUNDS).

3.3 Scoring and Incentives

Forecast accuracy was measured by mean squared error (MSE) across the scored rounds. At the end of each round, students were shown the running MSE for the scored rounds:

$$\text{MSE} = \frac{1}{T} \sum_{t=3}^T (\hat{\pi}_t - \pi_t)^2, \quad (2)$$

where T is the current round number, $\hat{\pi}_t$ is the student’s forecast, and π_t is the actual inflation rate for round t . Students learned in the instructions that the MSE penalizes over- and under-forecast errors symmetrically. Students were also explicitly informed that lower MSE indicates better performance.

Performance incentives were intentionally minimal. Completing the experiment was required to earn the participation credit for the activity that counted toward the student’s final grades. Forecasting performance did not affect grades for the activity. Students were given the choice to opt-in to compete to be listed on the class leaderboard under a name of their choosing. Students were asked to do their best.

3.4 Implementation

The experiment was conducted in an intermediate macroeconomics course at the University of California, Irvine in the spring of 2026. Total enrollment for the course was about 400. Students completed the activity independently over a weekend, accessing the experiment through individual participation links distributed via email. While this experiment was run remotely and asynchronously, it could also be done in-class if time permits. 344 students (86%) completed the experiment and 94.1% of completers consented to their data being used anonymously in this study.⁵

The post-experiment debrief is the pedagogical payoff of the experiment. After the experiment closed, students were shown the distribution of student forecasts alongside actual inflation for each round, the median forecast error over time, and the cross-sectional dispersion of forecasts. Having just experienced the difficulty of forecasting inflation, students are well-positioned to make sense of the professional forecasting data discussed in Section 2 and to appreciate why systematic forecast errors persist even among experts.

4 Results

Figure 7 plots the actual PCE inflation rate for the U.S. from 1970 to 1989 along with the individual forecasts of the students. In general, there is a clear central tendency to the inflation forecasts. Showing this graph to the students allows them to see how experiment participants who are presented with the same historical data will arrive at different forecasts, but in spite of the variation, a central tendency emerges that can be associated with the class's expectation for inflation. It is instructive to compare the students' forecasts of PCE inflation with the SPF forecasts of the GDP deflator in Figure 8. Like the experimental data, the professionals show considerable disagreement but also show a central tendency that appears to roughly track the actual data.

⁵This study was determined to be exempt from IRB review by the Institutional Review Board at the University of California, Irvine (STUDY00001417).

Figure 9 shows the actual PCE inflation rate for the U.S. with the median student forecast and the ± 1 standard deviation band. The median forecast appears to lag behind the actual inflation rate as students appear to be following a forecasting strategy that looks like adaptive learning. Student forecast errors show the same pattern of persistence seen in the SPF data: many consecutive years of either positive or negative forecast errors. Student forecasts are often within one standard deviation of the actual inflation rate except notably during the rise in inflation in the mid-1970s. For comparison, Figure 10 shows the median GDP deflator inflation forecast from the SPF, the ± 1 standard deviation band, and the actual inflation rate. Qualitatively, the SPF data aligns with the experimental results. Professional forecasts are often within one standard deviation of the actual inflation rate except when inflation rises suddenly, as in the mid-1970s.

While the data in Figures 7 and 9 and Figures 8 and 10 are based on forecasts of different series, it's still useful to quantify the quality of the forecasts. The third row of Table 1 reports the normalized RMSE of the SPF forecasts and the student experimental forecasts. The normalized RMSE accounts for the difference in the means of the two series. The professionals appear to be better at forecasting, but the quality of the student forecasts is comparable to the professionals. The result provides pedagogical justification for claims for assuming that the public forms expectations of inflation that are reasonable.

At the end of the experiment, students were asked to choose from four options to report how they made their forecasts. Results reported in Table 2. 55% claimed to use the three indicators provided when making their choices while 18% considered only the past values of the inflation rate. Of the 344 participants, 87 (25%) provided a written description of their forecasting strategy. Most students relied primarily on recent inflation as an anchor, adjusting for other indicators: "I mainly used recent inflation as the main indicator, then adjusted based on the unemployment and the federal funds rates." Others applied simple mean-reversion logic: "Once inflation gets really high like 10% it must go back down because of the FED." A notable minority

were candid about their approach: “vibes.” Encouragingly, some students connected the experiment explicitly to course concepts: “That was so fun! It made me understand how inflation expectations can be lagging the actual inflation in the short-run, but in the long-run, expected inflation will be around the same as the actual inflation rate.”

Students were also asked to report their effort level and their responses are summarized in Table 3. The results suggest that a little more than a quarter of the participants put in little to no effort which is reasonable given that there was no grade or monetary incentive.

5 Conclusion

I have described two complementary tools for teaching inflation expectations in intermediate macroeconomics: a ready-to-use dataset combining professional inflation forecast data with realized inflation and interest rate data and an online forecasting experiment that gives students direct experience with forecasting inflation in real time. Together, these resources allow instructors to give students an empirically and experientially grounded perspective on how inflation expectations are formed.

References

- Angeletos, George-Marios, Zhen Huo, and Karthik A. Sastry**, “Imperfect Macroeconomic Expectations: Evidence and Theory,” *NBER Macroeconomics Annual*, May 2021, *35*, 1–86.
- Atwood, Alicia, Tisha L. N. Emerson, Melissa A. Knox, and Mahjuja M. Taznin**, “Online platforms for classroom experiments: A primer for new adoptees,” *The Journal of Economic Education*, 2023, *54* (4), 382–390.
- Carter, Linda K. and Tisha L. N. Emerson**, “In-Class vs. Online Experiments: Is There a Difference?,” *The Journal of Economic Education*, January 2012, *43* (1), 4–18.
- Chen, Daniel L., Martin Schonger, and Chris Wickens**, “oTree—An open-source platform for laboratory, online, and field experiments,” *Journal of Behavioral and Experimental Finance*, March 2016, *9*, 88–97.
- Chen, Xiaofen**, “A Simple Classroom Experiment on Money Demand,” *Journal of the Scholarship of Teaching and Learning*, February 2018, *18* (1), 115–135.
- Cortés, Darwin, César Mantilla, and Laura Prada**, “Renewable resource dynamics: A Web-based classroom experiment,” *The Journal of Economic Education*, January 2023, *54* (2), 145–157.
- Duffy, John and Brian C. Jenkins**, “A classroom experiment in monetary policy,” *The Journal of Economic Education*, April 2019, *50* (2), 89–107.
- Ewing, Bradley T., Jamie B. Kruse, and Mark A. Thompson**, “Money Demand and Risk: A Classroom Experiment,” *The Journal of Economic Education*, 2004, *35* (3), 243–250.
- Hashimzade, Nigar, Oleg Kirsanov, and Tatiana Kirsanova**, “Lucas Islands game in intermediate macroeconomics,” *The Journal of Economic Education*, March 2026, *57* (2), 236–254.

Holt, Charles A., “Teaching Economics with Classroom Experiments: A Symposium,” *Southern Economic Journal*, January 1999, 65 (3), 603.

6 Tables

Table 1: Forecast performance of SPF professional forecasters and undergraduate students, 1970–1989. Mean forecast error (percentage points), root mean squared error (RMSE, percentage points), and correlation with realized inflation are computed across all individual forecaster-year (SPF) and student-round (students) observations. Normalized RMSE is RMSE divided by mean realized inflation over the sample period. SPF forecasts are of GDP deflator inflation; student forecasts are of PCE deflator inflation.

	SPF	Students
Mean error	0.66	0.19
RMSE	1.89	2.39
RMSE / mean inflation	0.32	0.42
Correlation	0.65	0.57

Table 2: Self-reported information use. Students were asked “Which information did you primarily use when forming your forecasts?” at the end of the experiment.

Description	Percent
Lagged inflation only	18.1
All three indicators*	55.0
Lagged inflation plus one other indicator	15.3
Some other approach	11.6

*Inflation, unemployment rate, and federal funds rate.

Table 3: Self-reported effort ratings. Students were asked “How much effort did you put into making accurate forecasts?” at the end of the experiment.

Rating	Description	Percent
1	I mostly clicked through without thinking	2.8
2	Some effort, but not much	25.9
3	Moderate effort	45.3
4	Fairly serious effort	17.5
5	I tried hard to forecast as accurately as possible	8.4

7 Figures

Figure 1: Median one-year-ahead GDP deflator inflation forecast from the Survey of Professional Forecasters and actual U.S. inflation (percent change in GDP deflator over the previous four quarters). Quarterly. Forecast values are lagged four quarters to align with the forecast horizon. Source: Survey of Professional Forecasters and FRED.

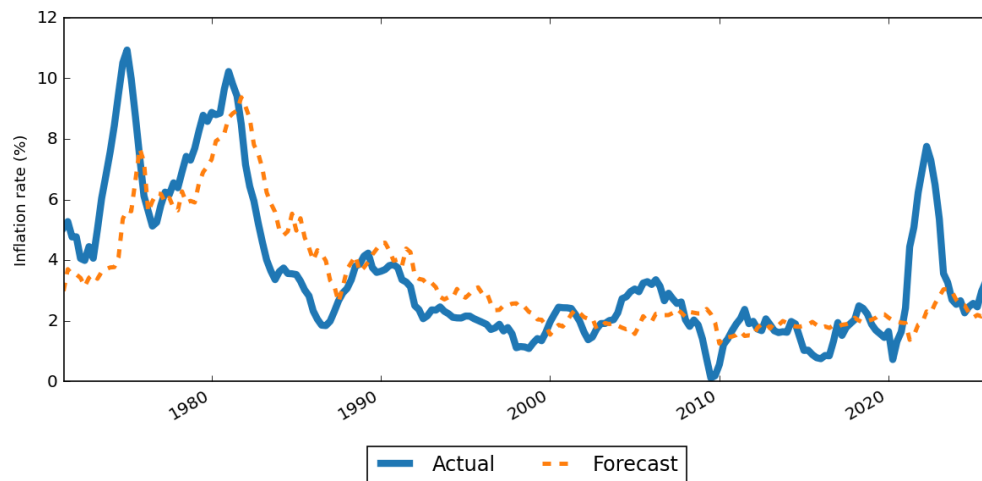


Figure 2: Median one-year-ahead inflation forecast from the University of Michigan Survey of Consumers and actual U.S. inflation (percent change in CPI over the previous 12 months). Monthly. Forecast values are lagged 12 months to align with the forecast horizon. Source: University of Michigan Survey of Consumers and FRED.

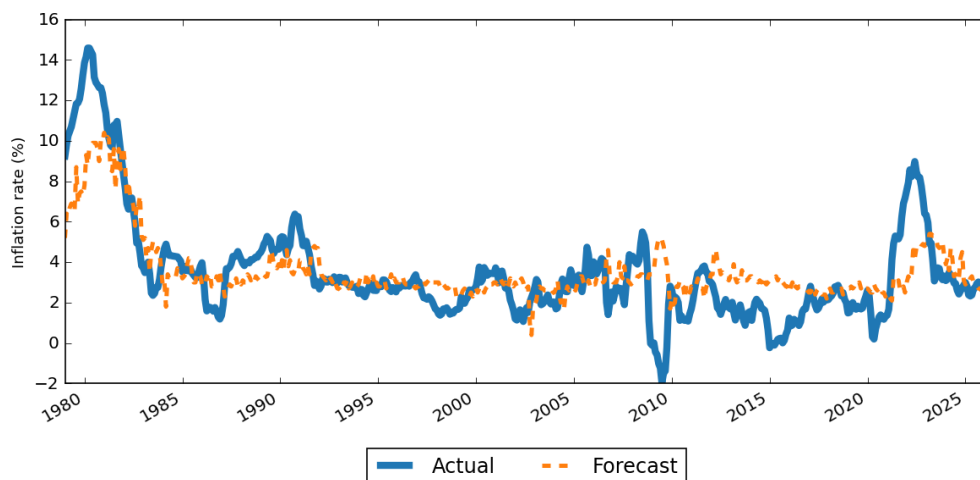


Figure 3: Median SPF one-year-ahead forecast and realized GDP deflator inflation, 1970–1989. Blue (red) shading indicates years in which actual inflation exceeded (fell short of) the median forecast. Shaded vertical bars denote NBER recessions.

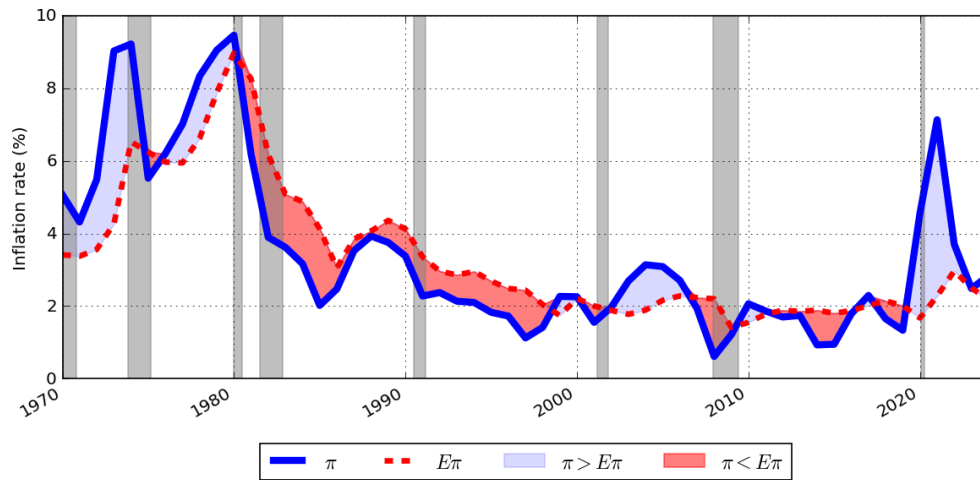


Figure 4: The ex ante real interest rate. Top panel: the nominal interest rate and the median SPF one-year-ahead inflation forecast. Bottom panel: the implied ex ante real interest rate, computed as the nominal rate less the median forecast. Shaded vertical bars denote NBER recessions.

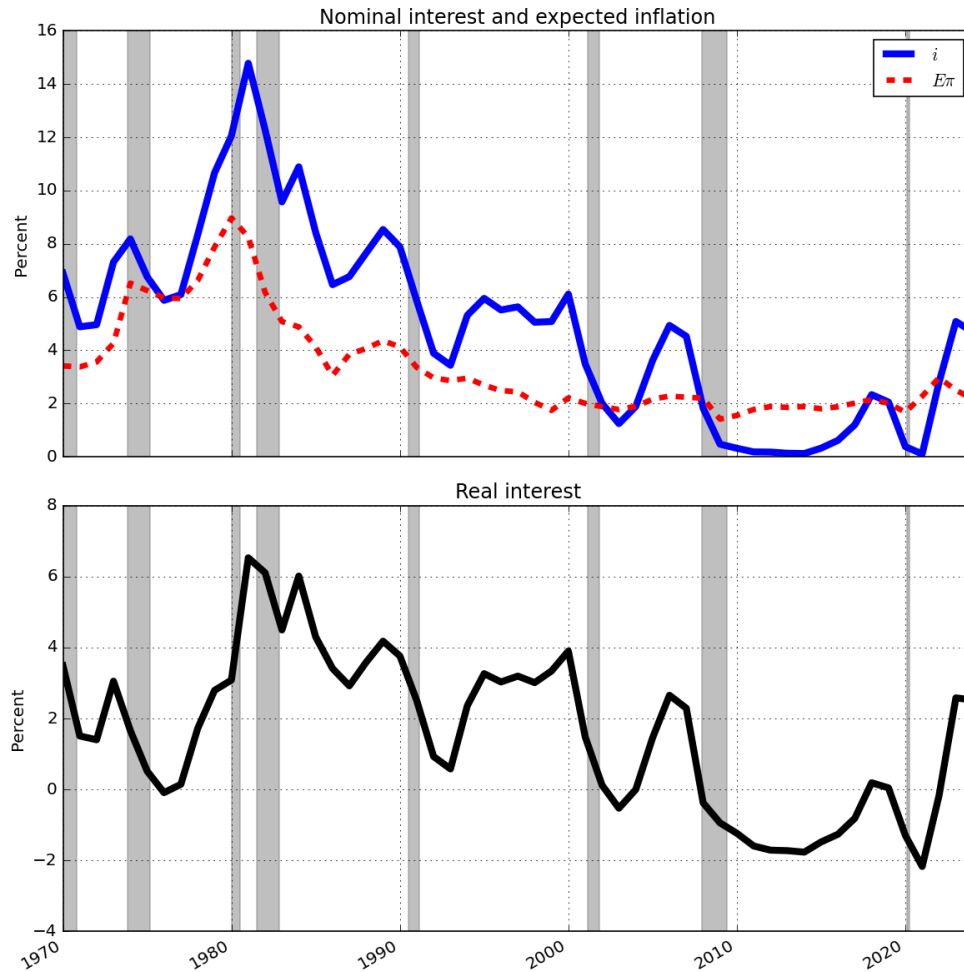


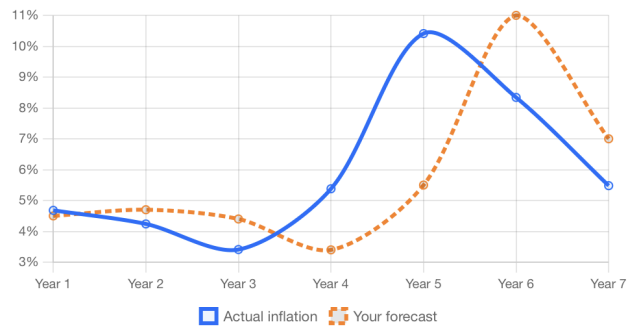
Figure 5: Decision screen shown to participants at the start of each round. Students were presented with a table of recent macroeconomic indicators and a chart showing actual inflation alongside their own prior forecasts, then asked to enter a forecast for the upcoming year.

Year 8

Time left to complete this page: 17:43

Recent macroeconomic data

Year	Inflation	Unemployment	Fed funds
Year 3	3.4%	5.6%	4.4%
Year 4	5.4%	4.9%	8.7%
Year 5	10.4%	5.6%	10.5%
Year 6	8.3%	8.5%	5.8%
Year 7	5.5%	7.7%	5.0%



Forecast inflation for Year 8:

Your forecast for next period's inflation rate (percent):

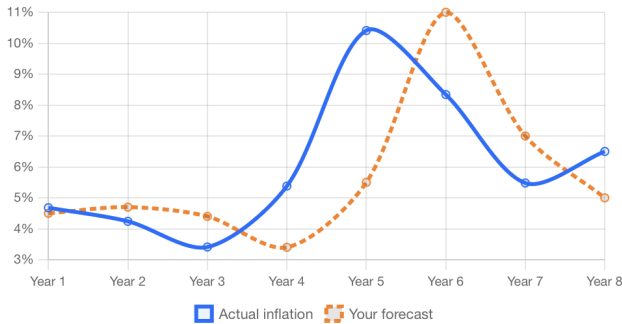
Next

Figure 6: Feedback screen shown to participants immediately after each forecast. Students were shown their forecast, the actual inflation outcome, the forecast error, the squared error, whether they over- or under-forecasted, and their running MSE across all scored rounds to date.

Feedback — Year 8

Time left to complete this page: 17:37

Your forecast	5.0%
Actual inflation	6.5%
Forecast error	-1.5%
Squared error	2.3
You	under-forecasted
Running MSE (scored rounds)	6.782



Next

Figure 7: Individual student forecasts of the PCE deflator inflation by round, 1970–1989, alongside realized inflation. Each grey line represents a single student.

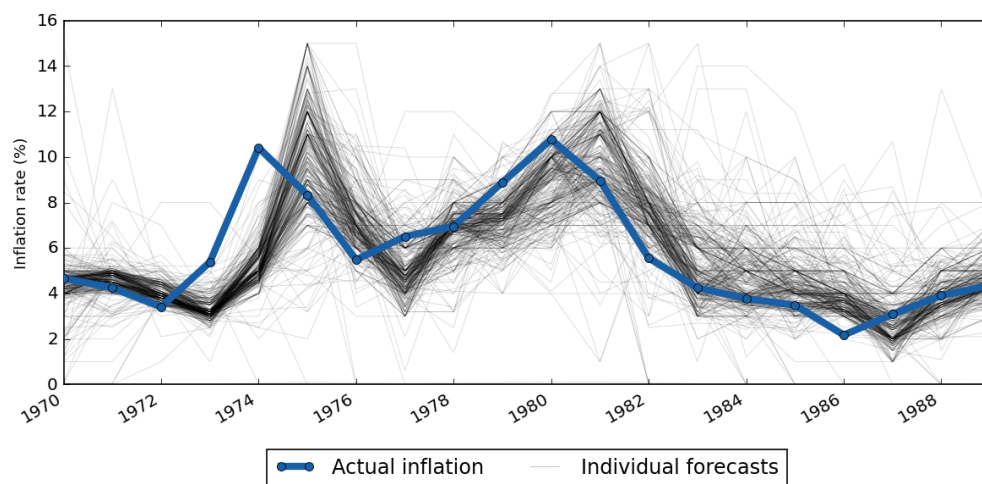


Figure 8: Individual SPF one-year-ahead forecasts of GDP deflator inflation, 1970–1989, alongside realized inflation. Each grey line represents a single forecaster.

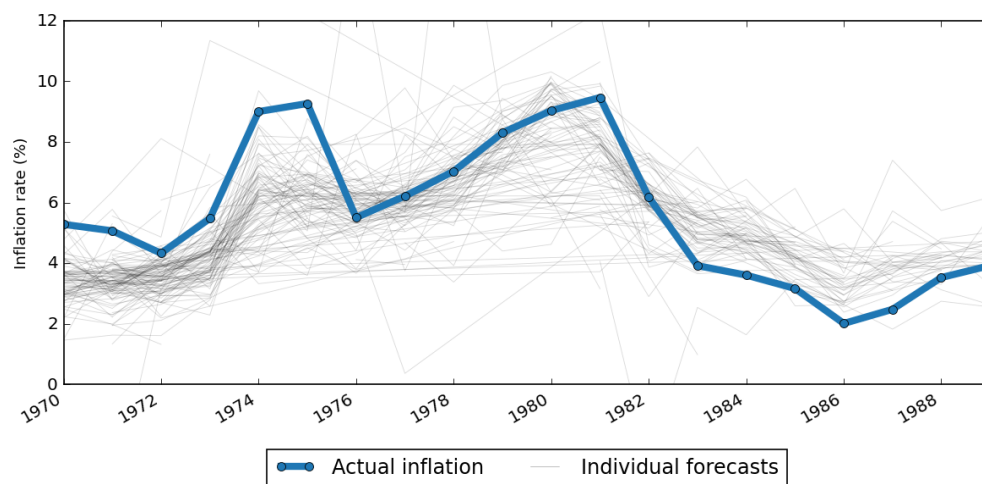


Figure 9: Median student forecast of the PCE deflator inflation and realized inflation by round, 1970–1989. Shaded band indicates ± 1 standard deviation of student forecasts.

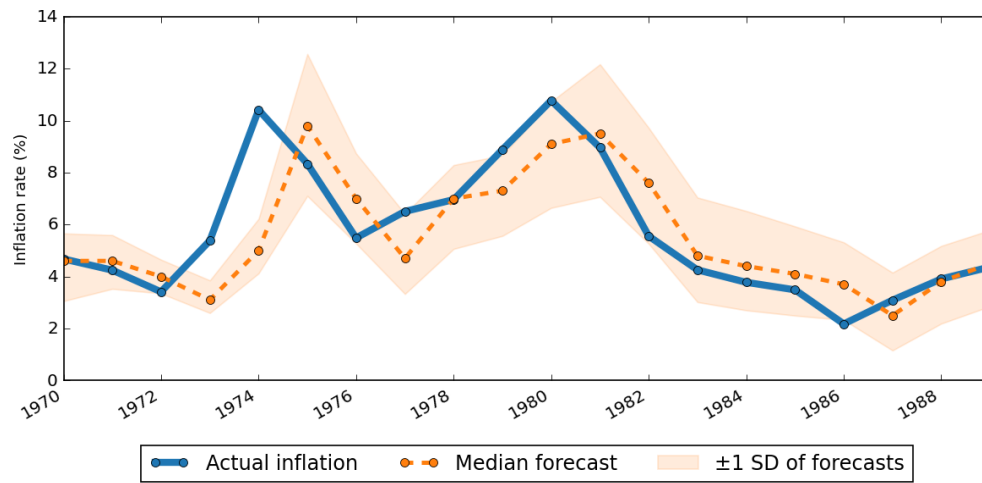


Figure 10: Actual U.S. GDP deflator inflation and the median SPF one-year-ahead forecast, 1970–1989. Shaded band indicates ± 1 standard deviation of individual forecasts.

